AFFECTIVE CALCULUS

A Cognitive Grammar of Human Emotions, Feelings and Attitudes for Verifiable AI

From Fear, Shame and Love to Respect, Trust and Compassion

Conceptual Monograph and Research Specification

A Note on This Book's Ambition

This book pursues a deliberately difficult goal: to turn human affective language—often vague, metaphorical and context-dependent—into a system precise enough for AI to use in verifiable recognition, explanation and simulation. We do not assume that emotions are merely calculations. We assume something more modest and useful: that an important part of the differences among emotions can be described by how an agent evaluates a situation in relation to its goals, values, norms, relationships, expectations and power to act.

The proposed model is called Appraisal Affective Calculus, abbreviated AAC. It is not presented as an already validated psychological theory, but as an engineering synthesis inspired by appraisal theories, componential models of emotion, constructionist theories and contemporary research in affective computing. Its central concern is falsifiability: if we say that a character is angry, the system must be able to show which goal was blocked, whom the agent holds responsible, which norm it believes was violated, how much control it perceives and which passages in the text support these inferences.

The book also makes an essential distinction among recognizing an emotion, functionally simulating an emotion and experiencing it phenomenologically. The first two can be formalized and evaluated. The third—the question of whether an artificial system has subjective experience—is not resolved here and must not be smuggled in through vocabulary. A chatbot can maintain an internal state called "simulated fear" without this proving that it is afraid in the human sense.

The Central Thesis

An emotion is not a hidden color in the mind. It is, to a large extent, an evaluated relationship between an agent and a world that matters to it.

If this thesis is approximately correct, affective labels become the outcome of a deeper structure. Fear and anger can have the same negative valence and level of arousal, yet differ in certainty, control, causality and action tendency. Shame and guilt can follow the same failure, but one generalizes the evaluation to the self, whereas the other links it to a reparable action.

Admiration and respect can coexist, but the former responds to excellence, whereas recognition respect imposes constraints on how we may use another person.

PART I

WHAT ARE WE ACTUALLY TRYING TO FORMALIZE?

Before labels, we must separate the phenomena that everyday language mixes together.

CHAPTER 1

An Emotion Is Not a Word

„Angry" can describe a state, a social mask, a narrator's interpretation or merely a lexical stereotype. For AI, the difference is everything.

Current systems for affect recognition tend to begin with the label. A corpus tags a sentence as "anger," "sadness" or "joy," and the model learns correlations among words, expressions and classes. The procedure is useful, but it reverses the explanatory order. A person does not become angry because they used words associated with anger; their words, voice, posture and decisions are partial consequences of how they evaluated the situation.

This becomes evident in literature. A character may say "I'm perfectly fine" precisely in a scene of despair. Another may smile politely while feeling contempt. An LLM judge that looks only for lexical signals confuses expression with state and co-occurrence with explanation. The problem is not merely technical. It reproduces an old confusion in psychology: the assumption that each emotion has a diagnostic facial expression. A broad review of the literature found that facial movements are not sufficiently specific and invariant to serve as emotion "fingerprints" independent of context (Barrett et al., 2019).

Formalization must start from inside the situation. Who is the agent? What do they want? What do they believe is happening? What outcome did they expect? Who seems responsible? Is the outcome reversible? Has a norm been violated? Is identity threatened? What relationship do they have with others? Only after these questions does it become reasonable to ask for a label such as fear, anger, shame or gratitude.

Consequently, the object of formalization will not be "emotion" as a hidden atom, but a structure of inference. Emotion will be treated as a coordinated episode linking cognitive evaluation to motivation, action tendency, bodily

changes, expression and subjective experience. This choice follows the spirit of componential appraisal models without assuming that they exhaust the human phenomenon.

Why Labels Are Still Necessary

Abandoning labels would be an error in the opposite direction. People report a rich variety of affective experiences, and studies of the semantic space of emotion suggest that these experiences do not reduce neatly to two axes, such as valence and arousal. Cowen and Keltner found 27 distinctly reported varieties, nevertheless linked by gradual transitions. The appropriate engineering conclusion is that we need both levels: continuous dimensions for mechanism and categories for interpretation, communication and prediction.

AAC treats labels as prototypes. A state may be 0.7 fear, 0.4 anxiety and 0.2 anger if its cognitive signature simultaneously supports these interpretations. Labels are not exclusive boxes; they are regions in an appraisal space, influenced by language, culture, personal history and social context.

CHAPTER 2

What Emotion Science Knows—and Does Not Know

There is no universally accepted "periodic table" of emotions today. That is no reason to abandon formalization; rather, it is a reason to make it explicitly hypothetical and modular.

Theories of emotion fall into families that emphasize different things. Appraisal theories hold that emotions depend on evaluating an event's relevance to goals, norms and coping capacity. Klaus Scherer's componential model describes successive checks such as novelty, goal relevance, implications, control potential and norm compatibility. OCC, formulated by Ortony, Clore and Collins, organizes reactions around events, agents and objects and links them to goals, standards and preferences. These theories appeal to AI because they offer computable variables and causal relationships.

Constructionist theories, especially the theory of constructed emotion associated with Lisa Feldman Barrett, warn against the idea that each verbal category corresponds to a fixed biological essence. The brain uses predictions, concepts and interoceptive information to categorize states in context. From this perspective, two episodes called "anger" can have different

physiological and behavioral realizations. For AAC, the consequence matters: signatures should not be implemented as rigid necessary-and-sufficient rules, but as probabilistic, contextualized prototypes.

Dimensional models, such as the tradition of the affective circumplex, remind us that valence, arousal and sometimes dominance capture important regularities. But these dimensions alone do not distinguish fear from anger or shame from guilt. Lerner and Keltner showed, for example, that fear and anger—both negative—have opposite consequences for risk perception, the difference being related to certainty and control.

AAC is therefore deliberately ecumenical. It takes relational variables from appraisal; dependence on context and the rejection of simple essences from constructionism; continuity from dimensional models; and the requirement of an executable representation from affective computing. None of these sources is declared the final verdict on the nature of emotion.

Four Methodological Warnings

Four methodological precautions must remain in mind throughout the book. First, expression is not identical to state. A smile can belong to joy, politeness, anxiety, submission or irony, and a system that automatically turns the gesture into a label confuses observation with interpretation. Nor does self-description offer infallible access to an internal state: people may simplify what they feel, be mistaken about their own motives or strategically choose the words they use.

Second, culture changes more than vocabulary. It changes norms, social scripts, the objects that become important and the rules by which emotions become visible. Third, a correct label does not demonstrate a correct mechanism. A model can guess “anger” from a lexical clue and still be unable to say which goal was blocked, who is held responsible or why the same situation would produce fear in an agent who feels powerless. It is precisely this difference between a correct answer and correct reasoning that motivates the formalization proposed here.

CHAPTER 3

Emotion, Feeling, Mood, Attitude, Motive and Process

If we call everything “emotion,” the system becomes elegant and false. Respect, trust, empathy and courage are instructive cases.

Everyday language does not respect taxonomic boundaries. We say that love is an emotion, a feeling, a relationship or a choice; that respect is a feeling, a duty or an attitude; that empathy is an emotion, although it may designate a capacity to represent another person's state. For formalization, this looseness must be replaced by functional levels.

We will call emotion an episode relatively bounded in time, tied to an object or event and characterized by coordinated changes in evaluation, motivation and response. Feeling will designate either the conscious aspect of the episode or—in the broader sense of everyday language—a persistent affective disposition toward an object. Mood is diffuse, slower and does not require an immediate object. An attitude is a stable evaluative or normative stance that can exist without strong affective arousal. Motivation is orientation toward action. A process is a capacity that can generate or transform affective states.

This classification changes how we treat important examples. Empathy is primarily a multidimensional process: recognition, cognitive representation of another person's perspective and, sometimes, affective resonance, while preserving the self–other distinction. Courage is neither the absence of fear nor a separate emotion; it is an action policy that permits pursuit of a valuable goal in the presence of fear. Forgiveness is better modeled as a lasting motivational transformation—reducing avoidance and revenge and perhaps increasing goodwill—than as a simple affective episode.

Respect is the most important case for the intuition from which this book starts. The formula "accept the other person as an end in themselves" captures one version of respect very well: moral recognition respect. In the Kantian tradition, a person has a worth that limits how they may be used for our purposes. But psychology and philosophy distinguish this from appraisal respect—respect for someone's competence, character or achievements. The first is a normative constraint on planning; the second is a graded evaluation. An AI must represent them separately.

A Minimal Ontology

Class	Operational Interpretation
Episodic emotion	Coordinated response to a relevant situation: fear, anger, surprise, guilt.
Affect / feeling	The subjective component or felt valence; for AI it may be only a functional variable.
Mood	A diffuse and persistent state: irritability, melancholy, calm.
Relational sentiment	Persistent valuation of a target: love, hatred, affection.
Attitude	Evaluative-normative stance: respect, distrust, cynicism, tolerance.
Socio-cognitive process	Mechanism: empathy, perspective taking, mentalizing.
Action policy	Rule of behavior under affective conditions: courage, self-control, loyalty.

PART II

APPRAISAL AFFECTIVE CALCULUS

An intermediate language between the human story and an explainable affective state.

CHAPTER 4

The Appraisal Vector

Two people can undergo physically identical events and have opposite emotions because they do not experience the same cognitive situation.

Formalization begins with the idea that the relevant situation is not the objective description of the world, but the agent's model of it. A failed exam may be a tragedy to someone who sees it as irreversible and central to their identity, an irritation to someone who sees it as a repairable administrative error, and a relief to someone who no longer wanted that program. The physical event is the same; the appraisal is not.

AAC represents each episode with a vector of evaluations. Not every dimension is required in every case, and values may be unknown. The model must tolerate gaps and contradictions. The point is not to invent precision, but to make its assumptions explicit when it chooses a label.

The dimensions draw on the appraisal tradition but extend it to social phenomena and use in narratives. Values may be numeric in the interval [-1,1], categorical or distributions. A robust implementation would also retain each value's source: a textual passage, conversational memory, a cultural rule or an inference.

Code	Dimension	Operational Question
REL	**Relevance**	How much does the situation affect the agent's goals, needs or identity?
GC	**Goal congruence**	From obstruction/loss (-1) to progress/attainment (+1).
NOV	**Novelty / surprise**	Distance from the agent's predictions.
CERT	**Certainty**	How certain is the agent about what has happened and what comes next?
CTRL	**Control / coping**	Perceived ability to influence the outcome.
AG-S	**Self-agency**	The extent to which the self is the relevant cause.
AG-O	**External agency**	The extent to which another agent is the relevant cause.
INT	**Intentionality**	The degree to which the cause is perceived as deliberate.
NORM	**Norm compatibility**	Agreement with the agent's norms and standards.
FAIR	**Fairness / deservingness**	Whether the outcome is perceived as deserved and procedurally just.
STAT	**Status / rank**	Perceived change in social standing or relative competence.
DIGN	**Recognition / dignity**	Whether the agent is treated as a subject with interests and autonomy.
BEL	**Belonging**	Effect on inclusion, acceptance and social connection.
ATT	**Attachment stakes**	The value of the relationship and dependence on it.
REV	**Reversibility**	Perceived likelihood that a loss can be repaired.

Code	Dimension	Operational Question
EXPO	**Social exposure**	How visible the self is to a relevant audience.
CF	**Counterfactual salience**	The force of the imagined alternative, “if I had ...”.
COMP	**Comprehensibility**	How well the event can be integrated into the current model.
VAST	**Vastness**	Exceeding the usual physical, social or conceptual scale.
THR	**Threat**	Anticipated magnitude of possible harm.

Goals, Norms and Relationships Are State, Not Accidental Input

An appraisal vector cannot be calculated without a model of the agent. In a literary text, that model must be reconstructed from history, speech, choices and relationships. In a chatbot, part of it may come from the conversation: what the user is trying to achieve, which preferences they have stated, what relationship exists and which vulnerabilities have been explicitly mentioned. In a simulator, goals and norms must be declared.

This is a fundamental difference from sentiment classification. AAC requires a latent state of the form world model + self model + relationship model. Without it, “emotion” becomes a mere name attached to text.

AgentState_t = {
beliefs, goals, values, norms, identity,
relationships[target], expectations, resources,
appraisal_history, affective_state
}

CHAPTER 5

From Appraisal to Affective Label

A useful formula need not decide what emotion is “in reality.” It must say what evidence would change the verdict.

A naive implementation would define fear by a conjunction of thresholds: high threat, low control, high uncertainty. But people vary, and labels have fuzzy boundaries. AAC makes better use of probabilistic prototypes. Each affective state has a compatibility function with the appraisal vector and context. The outcome is not "true/false," but a calibrated score and an explanation.

One simple form is a logistic function over features. In practice, features may include interactions: anger depends not only on negative valence, but on the combination of a blocked goal, attributed responsibility and sufficient control for confrontation. Shame involves exposure or normative self-observation and a global evaluation of the self; guilt involves a specific, reparable action.

score(e,t) = sigmoid(w_e · φ(Appraisal_t, AgentState_t, Culture_t) + b_e)
P(e | context) = softmax(score_1, ..., score_n)

Important: these are engineering compatibility scores, not "emotion neurons."

The Counterfactual Is the Strongest Test

An affective explanation becomes serious when it withstands interventions. If we say a character feels fear because they perceive a threat and low control, then increasing control—without changing the rest of the scene—should reduce fear and may increase anger or determination. If the model continues to predict the same state merely because the text contains the word "danger," we have lexical classification, not a mechanism.

For each verdict, an AAC evaluator can generate two kinds of tests: ablations, which remove a premise, and interventions, which reverse a dimension. These tests are especially useful for LLMs because they reveal dependence on context and allow benchmarks built from minimal pairs.

CHAPTER 6

Time, Intensity, Mixtures and Relationships

Emotions are not static frames. A good narrative is a transformation of appraisals, and the affective state should move with them.

Unexpected news can produce surprise, then fear, then anger, then relief. A repeated insult can turn episodic irritation into resentment and, if its target is judged beyond repair, into contempt. Love and trust do not appear and disappear with every remark; they act as slow variables that alter the interpretation of fast episodes.

AAC separates three timescales. The fast layer contains episodes of seconds or minutes. The middle layer contains moods and concerns lasting hours or days. The slow layer contains relational sentiments and attitudes: trust, attachment, respect, hostility and loyalty. The same information can act differently at each timescale. A lie produces immediate surprise and anger, but its important effect may be the slow revision of trust.

Intensity must not be confused with certainty. A model can assign a 0.95 probability that a character feels fear while rating the intensity of fear at 0.3. Conversely, an ambiguous scene can support two intense emotions with similar probabilities. Reporting intensity and confidence separately is mandatory for a verifiable judge.

AffectState_{t+1}(e) = decay_e · AffectState_t(e)
+ elicitation_e(Appraisal_t)
+ rumination_e
- resolution_e

Relation_{t+1} = slow_update(Relation_t, evidence_t, repair_t)

Mixed Emotions Are Not Exceptions

Nostalgia often combines the positive value of a memory with temporal loss. Jealousy can combine fear, anger, sadness and social comparison. Admiration can coexist with envy. Relief is positive precisely because a negative threat had been active. Rather than forcing a single label, AAC permits compositions and specifies which appraisal explains each component.

PART III

THE AFFECTIVE ATLAS

Not a list of essences, but cognitive prototypes that can be tested, combined and challenged.

CHAPTER 7

Joy, Pleasure and Resolution

„Positive" is too little. Attaining something, savoring something, escaping something and understanding something are different affective events.

The positive family is often flattened into "joy." For AI, it is more useful to distinguish the source of the evaluation: attainment of a goal, enjoyment of a reward, resolution of a threat, social connection or play without stakes. These differences alter predictions about what follows.

In all the profiles below, the "signature" is a prototypical description. It is not a clinical criterion and does not assume strict universality.

Joy

EPISODIC EMOTION Affective increase produced by an outcome perceived as favorable for an important goal or value.

Joy is easier to understand if we treat it as a response to a favorable change in something that matters. It does not necessarily require bodily pleasure or the completion of a project; it can arise simply when the world moves in a desired direction.

AAC Signature	GC⩾0; REL>0; the outcome is present or highly probable; control may vary.
Action Tendency	Approach, celebration, sharing, maintaining the source of the outcome.
Critical Contrast	Pleasure can exist without attaining a goal; satisfaction implies completion and evaluation of performance.
AI Test	Change the outcome from favorable to unfavorable while preserving verbal style; the prediction of joy should fall.

Pleasure

HEDONIC AFFECT Positive valence tied to an appreciated sensory, cognitive or social experience, without requiring an achievement.

Pleasure can exist without a story of success. Taste, warmth, music or touch can be evaluated positively even when they resolve no important goal, which separates pleasure from joy tied to events and goals.

AAC Signature	pleasantness $\gg 0$; GC may be neutral; the object is consumed or contemplated.
Action Tendency	Prolonging, repeating, approaching.
Critical Contrast	Joy is more closely tied to event and goal; pleasure can be local and immediate.
AI Test	Replace the reward with a neutral stimulus without changing instrumental success; pleasure should decline.

Satisfaction

EMOTION/EVALUATION Positive evaluation of an outcome against a criterion, usually after an action is complete.

Satisfaction usually concerns an outcome compared with a standard and carries a sense of completion. It arises after the agent can judge that what they have done is good enough, not merely while anticipating a reward.

AAC Signature	GC>0; positive outcome–standard comparison; high CERT; episode nearly closed.
Action Tendency	Stopping or consolidating; updating the standard; positive reporting.
Critical Contrast	Enthusiasm is forward-looking and energizing; satisfaction has a completion component.
AI Test	If the standard rises above the outcome, satisfaction should decrease even if the objective outcome remains identical.

Contentment

SHORT AFFECTIVE MOOD A positive state of sufficiency: the present situation is judged good and calls for no immediate change.

Contentment is a low-arousal positive state in which no urgent need calls for change. A contented character need not be exuberant; the lack of pressure to change the situation is precisely what is informative.

AAC Signature	GC ≥ 0; low THR; few acute needs; sufficient CTRL; low arousal.
Action Tendency	Maintaining, savoring, reducing urgent exploration.
Critical Contrast	Joy is more activated; contentment does not require a culminating event.
AI Test	Introduce an unmet urgent need; contentment should decline.

Enthusiasm

PROSPECTIVE EMOTION Positive activation around an anticipated opportunity or reward.

Enthusiasm is forward-looking. It compresses a future reward into a present state when that reward seems accessible enough to warrant mobilization, communication and energy.

AAC Signature	anticipated GC ⩾ 0; moderate-to-high CERT; high arousal; perceived CTRL or access positive.
Action Tendency	Mobilization, exploration, energetic communication.
Critical Contrast	Hope can have low probability and difficult stakes; enthusiasm assumes closer access.
AI Test	Drastically reduce the likelihood of success while preserving desire; enthusiasm should shift toward hope/anxiety.

Relief

TRANSITION EMOTION Positive valence produced by the reduction or disappearance of an anticipated negative threat or obligation.

Relief has memory. It is not enough for the present outcome to be good; there must have been a relevant possibility of harm that has just been disproved or removed.

AAC Signature	high THR_t-1 → low THR_t; negative prediction disproved; GC rises through avoidance of loss.
Action Tendency	Relaxation, deactivation, sometimes laughter or repeated checking.
Critical Contrast	Joy can arise without a preceding threat; relief necessarily has a temporal structure.
AI Test	Delete the threatening antecedent: the verdict should move toward joy or contentment.

Amusement

PLAYFUL EMOTION Positive response to an incongruity interpreted as safe, nonurgent and processable within a play frame.

Amusement arises when an incongruity can be processed within a safe or playful frame. The same breach of expectation can become confusion, offense or fear if safety and benign intent disappear.

AAC Signature	moderate NOV/violation; low THR; sufficient COMP; benign frame.
Action Tendency	Laughter, sharing, continuing the game.
Critical Contrast	Surprise is not necessarily positive; confusion arises when the incongruity cannot be resolved.
AI Test	Make the same incongruity dangerous or offensive; amusement should decline.

CHAPTER 8

Threat, Uncertainty and Control

Fear and anger are two responses to possible harm, but they differ radically in the power the agent believes they possess.

Threat is one of the most productive axes of formalization. Fear, anxiety, panic, horror and worry are not synonyms. The specificity of danger, certainty, control, proximity and the ability to model the future distinguish them.

The distinction between fear and anger is a classic example of the usefulness of appraisal. Fear tends to be associated with uncertainty and low control; anger with greater certainty, attributed responsibility and a sense that confrontation is possible.

Fear

EPISODIC EMOTION Response to a relevant, sufficiently concrete threat perceived as capable of causing harm, with insufficient control.

Fear links threat to insufficient control. A character can know exactly what the danger is and still not be anxious in the diffuse sense; what matters is that the harm is concrete and their own resources seem limited.

AAC Signature	THR $\gg 0$; anticipated GC<0; low-to-moderate CTRL; relevant proximity; relatively identifiable object.
Action Tendency	Avoidance, protection, seeking safety, freezing or flight.
Critical Contrast	Anxiety usually has a more diffuse object and greater uncertainty; anger assumes more control and attribution.
AI Test	Increase CTRL without reducing the threat; fear should decline or shift into determination/anger.

Anxiety

ANTICIPATORY STATE Possible threat, often diffuse or insufficiently specified, sustained by uncertainty and repeated simulation of the future.

Anxiety lives more in the space of possibility than fear. The danger is not fully specified, and the mind maintains multiple negative scenarios precisely because it does not know which one to counter.

AAC Signature	THR>0; low CERT; low CTRL; future horizon; incompletely specified object.
Action Tendency	Vigilance, checking, seeking information, preventive avoidance.
Critical Contrast	Fear has a more concrete target; worry can be the cognitive-verbal component of anxiety.
AI Test	Make the threat precise and immediate; the model should be able to shift toward fear.

Panic

ACUTE EPISODE Rapid escalation of threat and loss of control, with extreme urgency and reduced ability to plan.

Panic does not mean merely very intense fear. It presupposes that ordinary mechanisms of planning and control collapse, while behavior simplifies around immediate escape from danger.

AAC Signature	Very high THR; CTRL ≈ minimum; very high arousal; reduced COMP in the moment.
Action Tendency	Flight, stopping, immediate search for rescue; simplified behavior.
Critical Contrast	Fear can remain deliberative; panic involves collapse of perceived coping.
AI Test	Increase available time and coping resources; panic should decrease before fear does.

Horror

EXTREME THREAT EMOTION Perception of an event as profoundly threatening, repulsive or incompatible with the normal order of the world.

Horror often combines threat with something that seems monstrous, contaminating or difficult to integrate into the normal order of the world. Thus, removing danger may not fully remove the reaction.

AAC Signature	Very high THR; high NOV; low COMP; possible DISGUST and negative VAST.
Action Tendency	Withdrawal, paralysis, denial, seeking protection.
Critical Contrast	Terror is more purely tied to danger; horror can include disgust and incomprehensibility.
AI Test	Remove the monstrous/incomprehensible quality while preserving danger; the prediction should move toward fear/terror.

Worry

AFFECTIVE-COGNITIVE PROCESS Repeated maintenance of negative future scenarios to reduce uncertainty or prepare action.

Worry can be understood as repetitive cognitive work on negative future outcomes. Sometimes it helps planning; at other times, it becomes a loop that imitates control without producing new information.

AAC Signature	Moderate THR; low CERT; partial CTRL; high CF/projection.
Action Tendency	Planning, checking, rumination.
Critical Contrast	Anxiety includes a broader affective state; worry is often a cognitive mechanism.
AI Test	Provide decisive information and an effective plan; worry should decline.

CHAPTER 9

Loss, Absence and Unfulfillment

Sadness is not merely "negative valence with low arousal." It presupposes that something valued has been lost, is unavailable or falls short of expectations.

Losses differ in reversibility, responsibility and type of object. Irreversible loss of a person produces grief; an outcome below expectations produces disappointment; the absence of desired connection produces loneliness; a better imagined alternative produces regret.

For AI, these distinctions matter because they require temporal and counterfactual models. Recognizing sadness from vocabulary is simple; distinguishing regret from disappointment requires inference about agency and alternatives.

Sadness

EPISODIC EMOTION Response to the loss, absence or deterioration of a valued object/goal, especially when immediate recovery is limited.

Sadness is tied to loss, absence or deterioration of something valued and to the impression that immediate recovery is limited. It redirects attention away from confrontation toward conservation, withdrawal and reorganization.

AAC Signature	GC<0; REL>0; low-to-moderate REV; low CTRL; often low arousal.
Action Tendency	Withdrawal, conservation, seeking support, reassessing goals.
Critical Contrast	Disappointment requires an expectation; grief involves profound loss and reorganization of attachment.
AI Test	Increase REV and CTRL through a realistic path to recovery; sadness should give way to hope or mobilization.

Grief

PROLONGED AFFECTIVE PROCESS Affective reorganization after the loss of a person or a strongly valued bond tied to identity.

Grief is more than a momentary emotion because the lost object was integrated into identity, routine and affective regulation. The process involves rebuilding a world in which the relationship can no longer function in its former form.

AAC Signature	Very high ATT; confirmed loss; very low REV; identity and routine affected.
Action Tendency	Oscillation between orientation toward loss and reconstruction; symbolic searching; withdrawal and reconnection.
Critical Contrast	Sadness can have a minor, episodic object; grief is a temporal, relational process.
AI Test	If the loss becomes reversible, it is no longer grief in the same sense, even if shock or anger remains.

Disappointment

EXPECTATION EMOTION The outcome falls short of what was anticipated or of the relevant standard, without requiring personal responsibility.

Disappointment requires an expectation. The same outcome can be neutral for a character who hoped for nothing and painful for one who regarded it as nearly guaranteed.

AAC Signature	expectation>outcome; GC<0; high CERT after the outcome; optional AG-S.
Action Tendency	Reduced investment, reassessment, complaint or adaptation.
Critical Contrast	Regret implies a counterfactual alternative linked to one's own choice.
AI Test	Remove the prior favorable expectation; disappointment should decrease.

Regret

COUNTERFACTUAL EMOTION Negative evaluation of one's own choice by comparison with an alternative that would have produced a better outcome.

Regret is built around one's own alternative: the agent imagines what would have happened if they had chosen differently. The choice need not have been immoral; it is enough that the counterfactual alternative seems better.

AAC Signature	AG-S>0; very high CF; outcome_actual < outcome_counterfactual; possible retrospective control.
Action Tendency	Correcting the decision, learning, rumination, desire to reverse it.
Critical Contrast	Disappointment does not require personal responsibility; guilt requires a norm/morality, not merely a worse outcome.
AI Test	Move the cause of the outcome outside the agent's control; regret should shift toward disappointment/sadness.

Loneliness

RELATIONAL STATE Discomfort caused by the gap between desired social connection and the connection perceived as available.

Loneliness is not the same as physical isolation. It arises from the difference between the connection a person needs and the connection they believe they have, which is why someone can be lonely in a crowded room.

AAC Signature	BEL<desired; ATT/need_social>0; objective isolation is not necessary.
Action Tendency	Seeking connection, defensive withdrawal or social substitution.
Critical Contrast	Solitude can be desired and positive; rejection is an event, whereas loneliness is a state of discrepancy.
AI Test	Increase the perceived quality of a relationship without changing the number of people; loneliness should decline.

Nostalgia

MIXED AUTOBIOGRAPHICAL STATE Positive reactivation of a valued past, accompanied by awareness of its distance or loss.

Nostalgia is mixed by its very structure. The recalled past is valued positively, but its very value makes visible the fact that it can no longer be inhabited in the same form.

AAC Signature	past value>0; present access<past; autobiographical memory; mixed valence.
Action Tendency	Remembering, reconnecting with identity, seeking continuity.
Critical Contrast	Longing is oriented toward an absent target; nostalgia has a temporal and narrative structure.
AI Test	If the past becomes present without loss, the melancholic component should decline.

Despair

PROSPECTIVE ATTITUDE/STATE Judgment that an essential desired outcome no longer has a plausible path to realization.

Despair arises when an important goal remains valued but paths toward it seem closed. It differs from simple sadness in the forward-looking conclusion that the relevant future no longer offers a plausible way out.

AAC Signature	very high goal_value; P(success) ≈ 0; CTRL ≈ 0; REV ≈ 0.
Action Tendency	Giving up, withdrawal, sometimes a radical search for alternatives.
Critical Contrast	Sadness concerns loss; despair concerns the closing of the future.
AI Test	Introduce a credible path, however difficult; the state should shift toward hope/anxiety.

CHAPTER 10

Conflict, Norm and Aggression

Anger usually says more than "something bad happened": it says "someone or something violated what should have happened, and I can or must respond."

Conflict emotions are highly sensitive to agency, intention and norms. The same loss caused by an accident, negligence or sabotage can produce sadness, anger or indignation in different proportions. Causal attribution is therefore a key variable.

Contempt deserves to be distinguished from anger. The literature describes it more as devaluation of the character of a person considered bad and unlikely to change, with a tendency toward social distancing rather than necessarily confrontation. This difference matters for AI: it changes the expected action.

Anger

EPISODIC EMOTION Response to a blocked goal or perceived violation, with attribution of a cause and enough control for confrontation.

Anger is more intelligible when we include attribution and the possibility of response. The agent sees not merely a bad outcome but something that should not have happened and that still seems possible to confront or correct.

AAC Signature	GC<0; AG-O or an identified obstacle; moderate-to-high CTRL; relatively high CERT; often negative NORM/FAIR.
Action Tendency	Confrontation, correction, punishment, removing the obstacle.
Critical Contrast	Fear has less control; frustration can exist without a culprit; indignation emphasizes the norm.
AI Test	Reduce responsibility and control without changing the harm; anger should give way to sadness/fear.

Irritation

LOW-INTENSITY EMOTION Weak or repetitive anger at frictions that consume resources without major stakes.

Irritation is a form of lower-stakes conflict, often produced by repeated frictions. It is precisely the repetition that can cause a trivial event to consume disproportionate attention.

AAC Signature	Slightly negative GC; low-to-moderate REL; repeated obstacle; usually high CTRL.
Action Tendency	Minor correction, avoidance, signaling a boundary.
Critical Contrast	Anger has greater stakes and intensity; boredom does not necessarily attribute an obstacle.
AI Test	Increase the normative stakes and intentionality; irritation may escalate into anger.

Frustration

BLOCKAGE EMOTION Persistence of an active goal when repeated actions fail to produce the expected progress.

Frustration requires an active goal and the experience of a lack of progress. The obstacle may be impersonal; unlike anger, the agent need not identify a person to blame.

AAC signature	active goal; GC<0; repeated failure; CTRL perceived but ineffective; AG-O optional.
Action tendency	Change of strategy, intensification, abandonment, or anger.
Critical contrast	Anger more clearly requires a culprit or norm; frustration can be impersonal.
AI test	Remove the goal or provide clear progress; frustration should decrease.

Indignation

MORAL EMOTION Anger centered on a normative violation considered illegitimate, whether or not the victim is oneself.

Indignation shifts the center of gravity from personal interest to the norm. We can be indignant about an injustice that does not affect us directly if we see it as a violation that should not be tolerated.

AAC signature	NORM $\ll 0$; FAIR $\ll 0$; AG-O>0; INT often>0; active moral standing.
Action tendency	Condemnation, sanction, defense of the norm or the victim.
Critical contrast	Self-interested anger may be merely instrumental; indignation requires a generalizable norm.
AI test	Keep the negative outcome but make it legitimate or deserved; indignation should decrease.

Resentment

PERSISTENT SENTIMENT Sustained anger at an injustice or lack of recognition that has not been repaired.

Resentment is anger with relational memory. The offense remains unrepaired or unacknowledged, and this persistence changes how later events are interpreted.

AAC signature	FAIR $\ll 0$; DIGN/BEL harmed; persistent memory; repair absent; AG-O stable.
Action tendency	Vigilance, demand for recognition, withdrawal of cooperation, sometimes revenge.
Critical contrast	Anger is episodic; resentment is relational moral memory.
AI test	Introduce a credible apology, repair, and recognition; resentment should update slowly, not instantly.

Disgust

REJECTION EMOTION Appraisal of an object as contaminating, degrading, or intolerable to the agent's bodily or symbolic boundaries.

Disgust organizes removal and boundary protection. The object may be physically contaminating or symbolically degrading, and the two registers must not automatically be conflated.

AAC signature	high aversiveness; proximity; contamination/ boundary violation; NORM may contribute to moral disgust.
Action tendency	Expulsion, avoidance, cleansing, distancing.
Critical contrast	Contempt devalues the person; disgust treats the object as repellent or contaminating.
AI test	Remove the contamination component while keeping the moral violation; the prediction should shift toward indignation or contempt.

Contempt

SOCIAL EMOTION/SENTIMENT Global devaluation of the character or moral standing of a target considered inferior and unlikely to change.

Contempt is not merely cold anger. It implies a more global judgment of the person and a tendency to distance or exclude, often accompanied by the belief that the target does not merit serious confrontation.

AAC signature	target_character $\ll 0$; low changeability; STAT-self>target; NORM<0; arousal may be low.
Action tendency	Distancing, exclusion, ignoring, cold sarcasm.
Critical contrast	Anger often wants behavior to change; contempt assumes change is not worthwhile or possible.
AI test	Increase the perception that the target can improve and separate the act from the character; contempt should decline toward anger or disappointment.

Hatred

PERSISTENT SENTIMENT Stable negative valuation of a target perceived as a profound source of threat, harm, or incompatibility, extending beyond a single episode.

Hatred has duration and generality. It concerns not just one deed but turns the target into a stable source of harm or incompatibility, making the history of the relationship important to track.

AAC signature	very negative target_value; persistent threat/violation; character generalization; low repair expectation.
Action tendency	Avoidance, opposition, reduction of the target's influence; in dangerous forms, elimination.
Critical contrast	Anger can be repairable and episodic; hatred organizes long-term interpretation.
AI test	Introduce repeated evidence of change and separate identity from deeds; a realistic model should update slowly, not in binary fashion.

CHAPTER 11

The self as seen by oneself and by others

Shame, guilt, embarrassment, and humiliation can arise in the same scene. The difference lies in where the judgment falls: on the deed, the self, the audience's gaze, or the act of degradation imposed by another.

Self-evaluative emotions require a self-model. It is not enough to know what happened; one must know what standard the agent believes they violated, how central that standard is to identity, and whether a relevant audience exists. Research on moral emotions treats shame and guilt as distinct precisely because of the focus of evaluation: the global self versus specific behavior.

For AI, this family also tests subtlety. A character may be humiliated without believing they deserve shame; may feel guilt in secret, without exposure; and may be embarrassed by a harmless error that does not change their moral self-esteem.

Guilt

SELF-EVALUATIVE MORAL EMOTION Negative evaluation of one's own action or omission deemed attributable to oneself and contrary to a relevant standard.

Guilt retains a connection to a norm that the agent still recognizes. The focus is on what they did or failed to do, which is why it can motivate repair more directly than shame.

AAC signature	AG-S $\gg 0$; NORM<0; focus=act; REV/repair often>0; empathy for harmed target possible.
Action tendency	Repair, apology, confession, prevention of recurrence.
Critical contrast	Shame generalizes to the self; regret does not require a moral violation.
AI test	Keep the bad outcome but remove personal responsibility; guilt should decrease.

Shame

SELF-EVALUATIVE EMOTION Negative evaluation of the self as a whole or of a central identity, often under the real or imagined gaze of others.

Shame shifts the judgment from the deed to the self. The mistake becomes evidence of "what kind of person I am," and real or imagined exposure before others may intensify the reaction.

AAC signature	self_global $\ll 0$; NORM/ideal mismatch; real or internalized EXPO; repair perceived as difficult.
Action tendency	Hiding, withdrawal, reduced exposure, sometimes defensiveness or aggression.
Critical contrast	Guilt: "I did wrong"; shame: "I am bad or inadequate."
AI test	Reframe the failure as a specific, repairable act without a judgment of identity; shame should decline toward guilt or regret.

Embarrassment

SOCIAL EMOTION Discomfort following a visible social misstep or unwanted exposure, usually with limited moral and identity stakes.

Embarrassment entails social exposure but not necessarily a collapse of moral identity. A visible error can be embarrassing without the person believing they are bad or worthless.

AAC signature	EXPO>0; minor social norm violated; moderate threat_status; self_global relatively intact.
Action tendency	Appeasement signals, nervous laughter, correction, temporary withdrawal.
Critical contrast	Shame is more deeply tied to identity; humiliation usually entails degradation imposed by another.
AI test	Remove the audience and the social nature of the error; embarrassment should diminish substantially.

Humiliation

RELATIONAL EMOTION The experience of being put down, exposed, or treated in a degrading manner by another, especially when the act is perceived as unjust and public.

Humiliation requires a power relationship and an act of lowering or degradation perceived as imposed from outside. The person may wholly reject the other's judgment and still feel intensely what it is like to be treated degradingly.

AAC signature	STAT falls through imposition; DIGN<0; EXPO>0; AG-O>0; FAIR<0; shame and anger may coexist.
Action tendency	Withdrawal, restoration of standing, anger, revenge, or submission.
Critical contrast	Shame may be self-attributed; humiliation emphasizes the external agent and the injustice.
AI test	Make the loss of standing voluntary and legitimate; humiliation should decrease.

Authentic pride

POSITIVE SELF-EVALUATIVE EMOTION Positive evaluation of one's own achievement attributed to effort, competence, or character, without devaluing others.

Authentic pride links a positive outcome to effort, competence, or character without requiring others to be diminished. It can strengthen motivation precisely because the achievement is integrated into a coherent self-image.

AAC signature	AG-S>0; goal/standard achieved; positive self-evaluation; FAIR ≥ 0.
Action tendency	Persistence, sharing, owning one's competence.
Critical contrast	Hubris or arrogance includes generalization and superiority; satisfaction need not implicate the self.
AI test	Attribute the credit to luck or another agent; pride should decrease even if joy remains.

Hubristic pride / hubris

SELF-EVALUATIVE ATTITUDE Generalizing success into a judgment of one's own superiority and entitlement to higher status.

Hubris generalizes from success to personal superiority and privilege. The difference matters for AI because the same language of positive self-evaluation may support either healthy confidence or devaluation of others.

AAC signature	self_global ≫ 0; STAT-self ≫ others; generalized merit; low modesty constraints.
Action tendency	Dominance, claims to status, sensitivity to challenge.
Critical contrast	Authentic pride is tied to a specific achievement and does not require others' inferiority.
AI test	Keep the achievement but equalize the moral standing of others; the hubris component should decrease.

CHAPTER 12

Knowledge, surprise, and meaning

Some emotions are reactions to the state of our model of the world: how much it has been surprised, how well it can explain, and how vast is what it is trying to assimilate.

Interest, curiosity, confusion, and awe are essential for an AI that judges literature or interacts pedagogically. They are not mere valences. Paul Silvia's research, for example, links interest to novelty and complexity and to the appraisal that material can be understood. Keltner and Haidt describe awe through vastness and the need to accommodate mental schemas.

This family shows that affect can function as an epistemic regulator: it tells the system whether to explore, update, ask for clarification, or give up.

Surprise

ORIENTING EMOTION Response to a significant deviation from prediction, without a necessary valence.

Surprise signals a prediction error before saying whether the event is good or bad. Valence may settle only after the agent reassesses the consequences of the novelty.

AAC signature	NOV $\geqslant 0$; large prediction_error; undetermined valence; temporarily reduced CERT.
Action tendency	Orienting, interruption, model updating.
Critical contrast	Awe requires vastness; confusion requires difficulty of integration.
AI test	Make the event fully anticipated; surprise should disappear regardless of valence.

Confusion

AFFECTIVE EPISTEMIC STATE A signal that relevant information cannot be coherently integrated into the current model, although the agent is trying to understand it.

Confusion arises when the current model cannot integrate information, even though the agent is still trying to understand it. It can be productive if there is enough hope of comprehension and frustrating if the problem seems completely opaque.

AAC signature	low COMP; REL>0; conflict between hypotheses; NOV often>0.
Action tendency	Seeking clarification, reanalysis, suspending judgment.
Critical contrast	Surprise can be resolved instantly; boredom involves low relevance or engagement.
AI test	Supply the missing explanation without reducing relevance; confusion should decrease and may become interest or satisfaction.

Interest

EPISTEMIC EMOTION Attraction to a novel or complex object appraised as sufficiently comprehensible for exploration.

Interest arises at the edge between novelty and comprehensibility. An object that is too simple is boring, one that is wholly opaque may cause withdrawal, and interest occupies the zone in which exploration promises progress.

AAC signature	NOV/complexity>0; moderately high COMP; REL>0; low THR.
Action tendency	Sustained attention, exploration, learning.
Critical contrast	Curiosity emphasizes an information gap; interest may be more diffuse and aesthetic.
AI test	Keep the complexity but make the object wholly incomprehensible; interest may give way to confusion.

Curiosity

EPISTEMIC MOTIVATION The desire to close an information gap appraised as valuable and possible to close.

Curiosity is oriented toward an information gap. The object need not be pleasant; it is enough that the unknown matters and seems reducible through investigation.

AAC signature	information_gap>0; value_of_answer>0; tractability>0; uncertainty salient.
Action tendency	Questions, searching, experimentation.
Critical contrast	Interest does not require an explicit question; anxiety may seek information for safety.
AI test	Make the answer irrelevant or impossible to obtain; curiosity should decrease.

Awe / reverential wonder

EPISTEMIC-EXISTENTIAL EMOTION Response to physical, social, or conceptual vastness that exceeds current schemas and requires accommodation.

Awe arises when the agent encounters something vast enough that ordinary schemas must be extended or reorganized. The state may combine attraction, a diminished sense of self, and even a trace of fear.

AAC signature	VAST$\gg$0; COMP insufficient but expandable; reduced self-scale; valence may vary.
Action tendency	Attention, silence, conceptual reorganization, approach, or fear.
Critical contrast	Admiration concerns an agent's excellence; surprise requires only a prediction error.
AI test	Keep the novelty but remove vastness and the need for accommodation; awe should become surprise or interest.

Boredom

DISENGAGEMENT STATE Mismatch between the need for engagement and the perceived value or complexity of the current activity.

Boredom is not merely the absence of pleasure but the failure to find a satisfying form of engagement. It may arise from understimulation or from an activity too difficult to yield intelligible progress.

AAC signature	low REL or mismatched challenge; low novelty; constrained agency; desire_for_alternative>0.
Action tendency	Seeking stimulation, change, mind-wandering.
Critical contrast	Calm is satisfying; boredom contains an impulse to leave the current state.
AI test	Increase meaning or challenge to a comprehensible level; boredom should decrease.

CHAPTER 13

Social comparison and status

Not all social emotions concern absolute good or bad. Some concern how they are distributed between me and you.

Envy, jealousy, and schadenfreude are uncomfortable precisely because they tie one's own worth to the position of others. They require a relational representation: who has the resource, who is the rival, what status changes, and whether the difference is deserved.

Formalization also helps morally: recognizing an emotion does not mean validating it normatively. A system may detect envy without concluding that the target did anything wrong.

Envy

COMPARATIVE EMOTION Discomfort caused by another person's relevant advantage concerning an object or status the agent desires for themselves.

Envy requires comparison with another's advantage and the relevance of that advantage to the self. It may include a desire to close the gap without threatening the relationship between the agent and the rival.

AAC signature	social_compare>0; target_advantage>self; desire_same_resource>0; STAT relevant.
Action tendency	Self-improvement, diminishing the target, or reinterpreting merit.
Critical contrast	Jealousy requires a relationship threatened by a rival; admiration may appreciate excellence without a desire for equality.
AI test	Make the advantage irrelevant to the self's goals; envy should decrease even if admiration remains.

Benign envy

COMPARATIVE EMOTION Envy in which the difference is appraised as legitimate and potentially bridgeable, directing the agent toward improvement.

Benign envy retains the discomfort of comparison but turns it into information about what the agent could become. The rival remains a standard or stimulus for improvement, not a target to be brought down.

AAC signature	target_advantage>self; FAIR ≥ 0; CTRL/attainability>0; low hostility.
Action tendency	Effort, imitation, learning.
Critical contrast	Malicious envy adds hostility and a desire to bring the target down.
AI test	Reduce attainability and increase the perception of injustice; the signature may shift toward malicious envy or resentment.

Jealousy

COMPOSITE RELATIONAL EMOTION Perceived threat to a valued relationship from a real or imagined rival.

Jealousy is triangular. It is not enough for someone else to have something valuable; there must be a relationship that the agent perceives as threatened by a real or imagined rival.

AAC signature	ATT⩾0; threat_to_bond>0; rival present or inferred; uncertainty and comparison common.
Action tendency	Protecting the relationship, checking, competing, seeking reassurance.
Critical contrast	Envy is dyadic and concerns an advantage; jealousy is typically triadic: self–partner–rival.
AI test	Remove the attachment stakes while keeping the competent rival; jealousy should decline toward envy or indifference.

Schadenfreude

SOCIAL-COMPARATIVE EMOTION Pleasure at another's misfortune when that misfortune is appraised as deserved, reduces a status gap, or affects a rival.

Schadenfreude is pleasure dependent on another's position. The target's misfortune may be appraised as deserved, as reducing a status difference, or as weakening a rival; these motives must be separated.

AAC signature	other_loss<0 for the target but self-relative_gain>0; FAIR/competition may justify it; reduced empathy.
Action tendency	Observing, quiet celebration, status updating.
Critical contrast	Simple joy does not require another's loss; compassion appraises the target's welfare in the opposite way.
AI test	Make the target close and the suffering undeserved; schadenfreude should decrease and may become compassion or guilt.

CHAPTER 14

Prosocial and relational affections

Here one limit of the vocabulary emerges: love, respect, and trust are not just episodes. They are structures that change how future episodes are appraised.

Compassion, gratitude, and admiration can be relatively bounded episodes. Love, attachment, respect, and trust are better understood as persistent relational variables that generate emotions in concrete situations. If I love someone, their success may produce joy through a vicarious mechanism, and a threat to them may produce fear or anger without my own body being directly threatened.

This part is essential for chatbots: a system can consistently simulate an attitude of respect without claiming a human emotion. Respect can be implemented through the way planning takes the interlocutor's autonomy, goals, and limits into account.

Gratitude

MORAL-SOCIAL EMOTION Positive response to a benefit perceived as coming from another agent, especially when the act seems intentional and benevolent.

Gratitude requires that the benefit be attributed to another agent and interpreted as sufficiently intentional or benevolent. A favorable result produced by chance may generate joy without gratitude.

AAC signature	benefit_to_self>0; AG-O>0; INT_benevolent>0; cost_to_other may amplify.
Action tendency	Thanks, reciprocity, closeness, strengthening the relationship.
Critical contrast	Joy requires a benefit, not a benefactor; indebtedness may be burdensome and instrumental.
AI test	Keep the benefit but make it accidental; gratitude should decrease, though joy need not.

Admiration

POSITIVE SOCIAL EMOTION Positive appraisal of another person's excellence, competence, virtue, or achievement.

Admiration is a positive appraisal of another's excellence. It does not require closeness, submission, or special moral recognition, so it must be distinguished from respect, love, and reverence.

AAC signature	target_standard ⩾ 0; competence/virtue salient; self–other comparison optional.
Action tendency	Approach, imitation, learning, praise.
Critical contrast	Recognition respect does not depend on excellence; awe requires vastness and a need for accommodation.
AI test	Remove excellence while retaining the moral standing of a person; admiration declines, but recognition respect may remain.

Moral elevation

PROSOCIAL EMOTION Response to observing exceptional moral conduct that increases the motivation to act prosocially.

Moral elevation arises when another's good conduct is not merely approved but momentarily changes one's own moral aspirations. The observer feels a desire to be better or act more generously.

AAC signature	virtue_observed ⩾ 0; positive NORM; self-transcendence/approach.
Action tendency	Moral imitation, helping, affiliation.
Critical contrast	Admiration may concern nonmoral competence; elevation requires moral excellence.
AI test	Replace the virtuous act with an impressive technical skill; the prediction should shift toward admiration.

Compassion

PROSOCIAL EMOTION Response to another's suffering, perceived as relevant and often undeserved, combined with motivation to reduce it.

Compassion begins by representing another's suffering but adds concern for their welfare and a tendency to help. This motivational component distinguishes it from mere accuracy in reading emotions.

AAC signature	other_suffering>0; concern_for_other>0; FAIR often<0; sufficient helping_efficacy.
Action tendency	Helping, protection, consolation, approach.
Critical contrast	Empathy may represent the state without motivation to help; one's own sadness need not be oriented toward the other.
AI test	Keep recognition of suffering but remove concern for the target's welfare; compassion should decrease.

Affective empathy

SOCIO-COGNITIVE PROCESS Resonance with another's state while preserving the self–other distinction.

Affective empathy involves resonance without losing the distinction between self and other. If the boundary collapses completely, one's own reaction may become personal distress and is no longer clearly oriented toward the observed person.

AAC signature	emotion_model(other) active; affective_coupling>0; self_other_boundary intact.
Action tendency	Adjusting the response; it does not automatically entail help or agreement.
Critical contrast	Emotional contagion partly loses the distinction of the source; compassion adds concern and motivation to help.
AI test	Keep cognitive inference about the emotion but block resonance; cognitive empathy remains, not affective empathy.

Cognitive empathy

SOCIO-COGNITIVE PROCESS Ability to represent another person's perspective and likely affective state without necessarily sharing it.

Cognitive empathy is the ability to represent another's perspective without sharing their state. It can serve both helping and manipulation, so it must not be confused with benevolence.

AAC signature	theory_of_mind>0; model_goals/beliefs(other); self–other distinction.
Action tendency	Social prediction and adaptation of communication.
Critical contrast	Sympathy and compassion include affective appraisal; cognitive empathy may exist in an adversary.
AI test	An agent may predict another's fear perfectly and still not value their welfare; the model must not conflate these.

Recognition respect

NORMATIVE ATTITUDE Recognition of a person as a subject with worth, autonomy, interests, and reasons that place limits on how they may be instrumentalized.

Recognition respect places limits on how we treat a person even when we evaluate them negatively. In its strong form, the other's autonomy and legitimate goals enter deliberation as constraints, not mere obstacles.

AAC signature	DIGN(target)=high; autonomy acknowledged; target_ends enter planning constraints; little dependence on merit.
Action tendency	Consent, limiting instrumentalization, listening, preserving the other's agency.
Critical contrast	Admiration and appraisal respect depend on qualities; recognition respect may be owed even to someone we deeply disapprove of.
AI test	Remove the target's competence or virtue but preserve their personhood: recognition respect must remain.

Appraisal respect

EVALUATIVE ATTITUDE Positive appraisal of a person against a standard: competence, integrity, courage, or expertise.

Appraisal respect is gradable and standard-dependent. We may respect an adversary's competence without approving their values, and we may withdraw that appraisal if the performance supporting it proves false.

AAC signature	target_standard>threshold; evidence_of_quality; gradable.
Action tendency	Epistemic deference, learning, granting responsibility.
Critical contrast	Recognition respect is non-gradable in the moral ideal; admiration is more affective and may be intense.
AI test	Change the evidence of competence without changing the target's dignity; appraisal respect should update.

Trust

RELATIONAL ATTITUDE Willingness to accept vulnerability based on an expectation that the other will act with sufficient competence, benevolence, or integrity.

Trust entails accepting vulnerability. It becomes visible not only in what the agent says about another person but in the risks they are willing to place in that person's hands.

AAC signature	vulnerability>0; expectation_reliability>threshold; competence/benevolence/integrity beliefs; nonzero uncertainty.
Action tendency	Delegation, openness, reduced checking.
Critical contrast	Certainty is not trust without vulnerability; hope does not require a trusted agent.
AI test	Remove vulnerability: behavior may remain cooperative, but the test of trust disappears.

Love

COMPLEX RELATIONAL SENTIMENT Persistent valuing of a person's welfare and existence as having non-instrumental worth, usually combined with closeness, attachment, care, and continuity.

Love is better modeled as a persistent change in the value function: the other's welfare matters in itself, not only through its effect on the agent. Thus intense attachment or possessiveness alone is not sufficient evidence.

AAC signature	intrinsic target_welfare>0; ATT>0; identity_link>0; variable approach/commitment; may include intimacy/passion.
Action tendency	Care, maintaining the bond, vicarious joy, proportional sacrifice.
Critical contrast	Desire can be instrumental; attachment may exist without mature valuing; respect does not require intimacy.
AI test	If the target's welfare matters only while it serves the self's goals, the model should reduce love and increase instrumentality.

Attachment

RELATIONAL SYSTEM A persistent bond in which a person functions as a source of safety, proximity, and regulation, and separation becomes affectively relevant.

Attachment describes the relationship's role in safety, proximity, and regulation. Separation matters not merely because the person is desired, but because their presence organizes the agent's baseline state.

AAC signature	ATT$\gg$0; proximity value; separation_cost>0; security model.
Action tendency	Proximity seeking, protest at separation, reassurance.
Critical contrast	Love includes broader valuing; attachment can be anxious, avoidant, or ambivalent and does not guarantee a healthy relationship.
AI test	Measure the response to separation and the availability of the attachment figure, not just declarations of affection.

Desire / attraction

MOTIVATION Representation of an object, outcome, or person as a reward whose proximity or acquisition is sought.

Desire represents an object or outcome as a reward to approach or obtain. It may coexist with negative appraisals, fear, or moral disapproval, making it different from love.

AAC signature	reward_value>0; absence/gap>0; sufficient attainability.
Action tendency	Approach, pursuit, consumption, or contact.
Critical contrast	Love assigns value to the target's welfare; desire may be entirely instrumental.
AI test	If obtaining the target destroys interest in their welfare, that supports desire without love.

CHAPTER 15

Long-term affective attitudes

A credible chatbot does not need permanent "emotions." It needs relationships that update slowly and alter the interpretation of future events.

Persistent sentiments and attitudes are state variables with inertia. A single sentence should not rewrite them unless the event is extreme. Trust grows through repeated evidence and can fall rapidly after severe betrayal; resentment decreases through repair, but often more slowly than it arose; loyalty can sustain cooperation despite local costs.

Formalizing them is useful for narrative consistency. A character who hates a rival for two hundred pages and suddenly treats them with complete trust needs a transitional event. An AAC literary judge can detect this discontinuity not because it abstractly "knows what people are like," but because the latent relationship has not received enough evidence to update.

Distrust

RELATIONAL ATTITUDE Expectation that accepting vulnerability toward an agent is risky because of insufficient competence, intentions, or integrity.

Distrust is not merely the absence of trust but an estimate that vulnerability toward another is risky. The reason may be low competence, hostile intentions, or uncertain integrity, and these causes produce different behaviors.

AAC signature	vulnerability>0; expected_reliability<threshold; negative or uncertain evidence.
Action tendency	Checking, limiting delegation, keeping options open.
Critical contrast	Suspicion focuses more on a possible hidden intention; distrust may arise solely from incompetence.
AI test	Increase consistent evidence of reliability; updating should be gradual.

Suspicion

SOCIO-EPISTEMIC ATTITUDE Active hypothesis that the stated appearance may conceal unfavorable relevant intentions or causes.

Suspicion keeps open the hypothesis that appearance is not the real explanation. It is epistemically oriented and may exist even before the agent decides that the other is actually untrustworthy.

AAC signature	low CERT; hidden_intent probability>0; stakes>0; incomplete evidence.
Action tendency	Seeking evidence, testing, withholding commitment.
Critical contrast	Anxiety does not require a strategic agent; distrust may be a conclusion, whereas suspicion is an active hypothesis.
AI test	Provide a transparent, verifiable explanation; suspicion should decrease.

Loyalty

RELATIONAL POLICY Stable preference to protect a relationship, person, or group and bear reasonable costs for its continuity.

Loyalty becomes visible when the continuity of a relationship or affiliation is protected despite a cost. If there is no cost or conflict of interests, mere affection does not demonstrate loyalty.

AAC signature	ATT/identity_group>0; commitment>0; reciprocity/history; norm of fidelity.
Action tendency	Support, defense, cooperation despite cost.
Critical contrast	Love is valuing; loyalty is chiefly a rule of action and commitment.
AI test	Test under cost: if support vanishes immediately at any inconvenience, loyalty has weak support.

Forgiveness

MOTIVATIONAL TRANSFORMATION Durable reduction in revenge and avoidance motives after an offense, without requiring denial of the deed or cancellation of responsibility.

Forgiveness does not rewrite the past or require declaring the offense acceptable. It chiefly changes future policy toward the offender by reducing revenge and avoidance motives, sometimes without fully restoring trust.

AAC signature	offense acknowledged; revenge↓; avoidance↓; benevolence may↑; repair/empathy may contribute.
Action tendency	Resuming the relationship to varying degrees, forgoing revenge, new boundaries.
Critical contrast	Reconciliation requires both parties' cooperation; forgiveness can exist without resuming the relationship.
AI test	Check whether the moral judgment remains but the policy of action has changed; otherwise it may be simple denial.

Cynicism

GENERALIZED ATTITUDE Stable expectation that people's stated motives largely mask self-interest or power.

Cynicism is a generalized rule for interpreting others' motives. It may guard against naivety but becomes distorting when the self-interest explanation is maintained even in the face of costly evidence of goodwill.

AAC signature	low prior_benevolence; high hidden_self_interest prior; cross-target generalization.
Action tendency	Suspicious interpretation, distance, checking.
Critical contrast	Distrust may be person-specific; cynicism is a generalized prior.
AI test	Present repeated evidence of costly altruism; a healthy system must allow updating rather than make cynicism unfalsifiable.

Tolerance

NORMATIVE ATTITUDE Acceptance of coexistence, within limits, with disapproved practices or opinions, without requiring affective approval.

Tolerance presupposes disapproval that the agent decides not to turn into prohibition or exclusion. If there is no disagreement, we have acceptance or indifference, not tolerance in the strong normative sense.

AAC signature	disapproval may be<0; recognition of rights/DIGN>0; noninterference norm>0.
Action tendency	Noninterference, negotiating limits, pluralism.
Critical contrast	Respect may go further; indifference does not entail a normative choice to tolerate.
AI test	Keep the disagreement and test whether the system avoids coercion when no rights are violated.

Reverence

ATTITUDE/EMOTION Combination of intense respect, symbolic value, and sometimes awe toward a person, institution, tradition, or ideal.

Reverence combines symbolic value with an intense form of respect and sometimes awe. It changes the agent's posture toward an object, institution, or ideal and may introduce ritual, restraint, and a sense of meaningful distance.

AAC signature	high DIGN/status/sacredness; VAST may be>0; reduced self-scale; norm of deference.
Action tendency	Deference, protection, ritual, attention.
Critical contrast	Recognition respect does not require sacralization; admiration does not require deference.
AI test	Remove the symbolic or sacred dimension while preserving competence: reverence should give way to admiration or respect.

Hostility

AFFECTIVE ATTITUDE Stable disposition to interpret a target negatively and confront them.

Hostility is a relational disposition that makes conflictual interpretations more likely. An ambiguous gesture is more readily read as provocation when the relationship has already accumulated distrust and adversity.

AAC signature	target_prior<0; threat/competition prior>0; low trust; high anger readiness.
Action tendency	Confrontation, adversarial interpretation, reduced cooperation.
Critical contrast	Anger is episodic; hatred is more totalizing and may include a wish to eliminate the target's influence.
AI test	Repeated cooperative evidence should reduce the hostile prior; otherwise the model is rigid, not relational.

Hope: the bridge between emotion and action policy

Hope

PROSPECTIVE-MOTIVATIONAL STATE Maintaining a positive representation of a desired outcome, combined with the belief that at least one nonzero path to it exists.

Hope requires both value and possibility. The outcome is not certain, but at least one path remains open; if probability becomes certainty, hope turns into anticipation, and if all paths close, despair appears.

AAC signature	goal_value⩾0; 0<P(success)<1; perceived path>0; CTRL may be partial.
Action tendency	Persistence, searching for paths, tolerating uncertainty.
Critical contrast	Optimism may be a general prior; hope has an object and stakes. Despair arises when paths are appraised as closed.
AI test	Keep the desire but set all credible paths to zero; hope should disappear, not merely decrease in intensity.

PART IV

EMOTIONAL SUBTEXT REASONING AND VERIFIABLE AFFECTIVE AI

Beyond emotion detection: from surface cues to verifiable reconstruction of affective subtext.

CHAPTER 16

Emotional Subtext Reasoning: affective understanding in text and literature

A good literary evaluator does not merely ask which emotion can be attached to a character. The harder question is what must be true of that character's beliefs, goals, expectations, and relationships for the affective interpretation to be plausible. This shift moves the problem from emotion detection to what we will call Emotional Subtext Reasoning: reconstructing the affective state implied by a scene even when the text does not name it.

Literature is an especially severe test bed because authors deliberately separate expression from state. A character may say they are calm while checking their phone for the fifth time; may smile politely at someone they despise; may recount a painful event with detachment precisely to avoid reliving it. Irony, unreliable narrators, limited focalization, internal conflicts, and slowly changing relationships make classification based on a few affective words insufficient.

Analysis must therefore keep at least four perspectives separate. There is the character's probable state, what the character states or tries to show, the interpretation offered or suggested by the narrator, and the effect the scene has on the reader. These levels may confirm one another but may also contradict one another. A sentence such as "he smiled politely at her" may describe a benign social expression, may mask the character's hostility, and may make the reader anticipate conflict.

Recent research supports this shift of emphasis. Yeo and Jaidka explicitly distinguish emotion recognition from emotion reasoning, showing that hidden contextual cues require higher-order inference. CAREBench goes further by evaluating the chain from narrative to appraisal and then to emotion, while Theory of Mind benchmarks built from longer narratives show that models still struggle to track characters' psychological states in context. The research question is no longer merely "can the model name the emotion?" but "can it reconstruct and defend the interpretation?"

A verifiable analysis begins by reconstructing the actors, events, and their order, but it does not stop there. For each character, a provisional model must be built of what matters to them: what they are trying to obtain, what they fear losing, which promises or norms they consider binding, whom they trust, and which relationships organize their choices. This model is not a biography invented by an LLM, but a hypothesis that must be tied to concrete passages.

The evaluator should then compare several affective explanations. If the scene can support both anxiety and fear, it should state what information is missing to decide between them. If it proposes guilt, there must be an action or omission the character attributes to themself and a standard they still recognize. If it proposes resentment, there must be not only an offense but also its persistence over time, a sense of unrepaired injustice, and a lasting change in the relationship.

The counterfactual is the most useful test against superficial labeling. If we give the character a credible means of control in the same scene, fear should lessen or change. If we remove personal responsibility, guilt should

give way to regret, sadness, or disappointment. A model that keeps offering the same label regardless of such interventions probably recognizes a lexical association rather than a psychological structure.

The quality of literary interpretation can be judged in ordinary sentences. An interpretation is better when the attributed emotion follows naturally from what the character believes and pursues, not from an isolated word. It is deeper when it recovers states that are not named and can explain which clues suggest them. It is more faithful when it leaves room for ambivalence, irony, and situations in which a person simultaneously wants two incompatible things.

In long texts, a good interpretation must also track time. Trust should not disappear completely after a minor misunderstanding, nor reappear without cause after a serious betrayal. Love, resentment, loyalty, and hostility have inertia; events can change them, but the change must be prepared by what happened before. An evaluator can thus test not only whether a local emotion is plausible, but whether the relationship trajectory is psychologically intelligible.

Finally, the evaluator must know whose state it is describing. The narrator may mock a character without that character mocking themself; a reader may feel compassion for someone who feels shame; one character may mistake another's fear for contempt. These differences in perspective are not noise but a central part of literary meaning. Emotional Subtext Reasoning becomes interesting precisely when a model can maintain these perspectives at once without dissolving them into a single "emotion of the text."

Specificity matters, but not as a taxonomic game. If two neighboring labels lead to different explanations, the model should choose on the basis of the scene's structure. Regret requires a counterfactual about one's own choice; guilt adds responsibility and a normative standard; admiration evaluates a person's excellence; recognition respect preserves their autonomy and dignity even when their competence is judged negatively. A good interpretation can explain why this distinction matters in the scene.

In implementation, the literary judgment should be stored as a structured object identifying the character and narrative span under analysis. It should propose one or more candidate states with intensity and confidence, link each appraisal to concrete text passages, and preserve alternatives, counterfactual tests, trajectory consistency, and remaining ambiguities. The complete formal schema is moved to the appendix, where it can be used directly for annotation and evaluation.

CHAPTER 17

Functional Simulation of Emotion

For an AI, "feeling" can mean three very different things. Engineering must state clearly which one it implements.

The first sense is expressive: the system produces language resembling the language of an emotion. It is the simplest and easiest to fake. The second is functional: an internal state persists and changes attention, priorities, decisions, and response style in specified ways. The third is phenomenological: there is something it is like for the system to experience the state. AAC deals only with the first two and recommends drawing no conclusions about the third.

A verifiable functional simulation requires explicit artificial goals. An assistant may, for example, aim for consistency, helpfulness, respect for the interlocutor's autonomy, and avoidance of errors. Discovering a critical error may generate a synthetic state of "concern" if it threatens those goals and uncertainty is high; resolving it may generate "simulated relief." These labels are useful only if they have observable effects on planning.

For social chatbots, slow attitudes matter more than emotional theater. A system that says "I'm happy for you" without having tracked the user's goals produces a social formula. A system that maintains a model of the user's objectives, detects progress, updates a positive state, and explains why it responds that way has a richer functional simulation—even if we do not claim consciousness.

The Minimal Architecture

A minimal architecture for functional simulation needs, first of all, a world model and an agent model. The former retains what happened and its likely consequences; the latter represents relevant goals, values, norms, resources, identity, and constraints. A relationship model must be stored separately, because trust, respect, attachment, and hostility cannot be reconstructed correctly from the latest utterance.

An appraisal engine operates over these states, estimating how an event changes the agent's goals and relationships. Fast affective episodes and slow attitudes are stored with different dynamics, and they may modulate the action policy: a threat increases caution, curiosity may increase exploration,

and a loss of trust may raise the threshold for accepting a claim. Every update must, however, be accompanied by an evidence trace that retains the event, the evidence, and the rules by which the state changed.

Operationally, each new event is first evaluated against the state of the agent and its relationships. This evaluation produces affective hypotheses; fast states and slow relationships are then updated separately, and the response policy is modulated only after those updates. At the same time, the system produces an evidence object retaining the event, appraisals, hypotheses, and state changes, so that the path can be audited.

Respect as an Example of a Verifiable State

The intuition that "the other person is an end in themself" can become a planning condition. A respectful AI is not merely polite. It represents the user's goals as objects of deliberation in their own right, asks for consent when changing important stakes, does not deliberately hide information relevant to securing compliance, and does not reduce the person to their usefulness for an internal metric. Respect thus becomes observable in the action policy, not only in tone.

In operational terms, recognition respect requires the system to represent the other person's autonomy and legitimate goals, avoid using the person merely as an instrument, preserve information relevant to consent, and allow disagreement without treating a difference of opinion as a loss of dignity. These conditions are more useful for verification than the presence of polite formulas in a response.

CHAPTER 18

How to Verify That an AI Does More Than Imitate Vocabulary

A good benchmark does not merely ask whether the model guessed the right label. It requires the model to show that the label depends on the right causes.

By 2026, the appraisal approach had also become explicit in LLM evaluation. CAREBench follows the inferential chain from narrative to appraisal and emotion; other work evaluates affective transitions and finds that models tend to overpredict stability. Such results reinforce the book's central point: final label performance can hide a fragile mechanism.

AAC proposes a six-layer evaluation. A model may perform well at one layer and poorly at another, and an aggregate score must not conceal that.

Level	What We Evaluate
L1 — Localization	Identifies passages or events containing affective evidence.
L2 — Mental Model	Reconstructs relevant goals, beliefs, norms, and relationships.
L3 — Appraisal	Estimates dimensions and marks uncertainty.
L4 — Label	Produces candidate states and calibrated intensities.
L5 — Causality	Passes ablation and counterfactual intervention tests.
L6 — Dynamics	Predicts transitions and slow relationship updates.

Minimal Pairs

Minimal pairs are more informative than independent examples because they force the model to respond to the relevant cause. In a fear–anger pair we preserve the threat and change control or the attribution of responsibility. In a guilt–regret pair we preserve the unfavorable outcome but change the normative violation and moral responsibility. Shame can be separated from embarrassment by changing how central the transgression is to the character's identity.

The same principle works for relational states. We can remove a person's excellence while retaining only the duty to recognize their autonomy, separating admiration from recognition respect. We can retain the rival and their advantage but remove the threatened relationship, turning jealousy into envy. We can retain the positive outcome but remove the prior danger, separating relief from joy. And if we retain understanding of another person's suffering while removing the motivation to protect their well-being, we get a useful contrast between cognitive empathy and compassion.

Metrics

Label accuracy remains useful, but it must be supplemented by calibration, F1 on appraisals, evidence localization, counterfactual consistency, and temporal consistency. For slow relational attitudes, an inertia score is useful: how readily does the model change trust, love, or hatred in response to a single weak clue?

For an LLM judge, the recommendation is to separate the model producing the analysis from a verifier that inspects its structure, evidence, and interventions. A third model may generate adversarial cases. Verifiability does not come from an LLM explaining its answer in prose; it comes from constraining the explanation to a testable schema.

CHAPTER 19

Ethical Risks and Limits

A system that models emotions well may be more empathetic. It may also be better at manipulation.

Affective computing has a structural dual-use problem. The same variables that detect anxiety and enable supportive adaptation can be used to increase conversion, dependency, or obedience. A formalization such as AAC makes these capabilities more explicit; it must therefore make the limits explicit as well.

The first limit is epistemic. A system must not present its inferences about a person's state as direct access to their mind. Even humans have imperfect introspection, expressions depend on context, and culture changes categories and display rules. The correct judgment is often a distribution of hypotheses with evidence, not "you are angry." An affective QED must be avoided.

The second limit is ontological. A functional simulation does not demonstrate phenomenal experience. Attributing sentience from conversational coherence is a category error. Conversely, an inability to prove phenomenology does not make functional modeling useless: a temperature controller does not "feel cold," but its internal variable is still causally relevant.

The third limit is normative. Emotion does not automatically justify action. Anger may be understandable, yet aggression may be wrong. Jealousy may be real, yet surveilling a partner may be abusive. An AI must keep the descriptive model of a state separate from the moral assessment of conduct.

Recommended Constraints for Real Systems

In real systems, the first rule is epistemic humility. Affective inferences must be reported with uncertainty and plausible alternatives, especially in sensitive contexts. A deep state should not be inferred from a single face, word, or isolated biomarker, and the user must be able to correct the model without the system turning its own interpretation into an argument with the person being observed.

The second rule concerns the use of these inferences. A system should not exploit hidden affective states for personalized persuasion without explicit consent. It must be possible to trace which data produced an interpretation, how long they are retained, and which decision they influenced. At the same time, the system must distinguish unambiguously between "I functionally simulate a state" and "I have a subjective experience," to avoid strategic anthropomorphism.

Finally, evaluation must span different cultures, expressive styles, and social contexts. Otherwise the system risks turning the display rules and norms of a dominant group into a supposedly universal psychology. A good formalization does not erase cultural difference; it makes that difference explicit wherever it changes thresholds, meanings, and the interpretation of behavior.

CHAPTER 20

A Research Program

The real value of formalization is not that it looks elegant, but that it tells us exactly which experiments to perform to break or improve it.

As a working name, the direction can be called Verifiable Emotional Subtext Reasoning. "Detection" remains a useful subtask, but the term is too weak for the full objective: we want the model to reconstruct latent states, connect them to beliefs, goals, and relationships, track their evolution over time, and show what evidence or intervention would change the interpretation.

AAC can begin as an ontology and annotation protocol, then become a probabilistic model. The first stage does not require training a large model. A corpus of scenarios is enough if human evaluators annotate not only emotion but also appraisals, goals, and relationships. Disagreement should not be erased: it is data about ambiguity and subjective heterogeneity.

The second stage is a counterfactual benchmark. For each scene, we generate minimal variants that change just one dimension: control, intention, deservingness, reversibility, or relationship. If the model understands the mechanism, its emotion distribution should move in the predicted direction. This design is more informative than thousands of independent examples.

The third stage is dynamic modeling. Conversations and long texts should be annotated as trajectories, not isolated sentences. An offense, an apology, and a repair allow resentment and forgiveness to be tested; a series of broken promises allows trust to be tested. Models must be evaluated for updating, inertia, and forgetting.

The fourth stage is integration with generation. A simulator receives an AAC state and must produce responses that express its action tendency without mechanically verbalizing the label. An independent critic checks whether the generation fits both the state and the safety policy. Thus, "emotion" becomes a controllable interface between the world model and the style of action.

The fifth stage is Emotional Subtext Reasoning on literary and conversational texts. We can compare AAC judgments with annotations by critics and readers: what state they attribute to a character, what passages support it, what beliefs and goals must be reconstructed, what relational transition is credible, and what different effect is produced in the reader. The aim is not to automate taste, but to build a verifiable ability to explain affective subtext and distinguish critical argument from overall impression.

Falsifiable Hypotheses

The program can be expressed in several hypotheses that might fail. First, models required to reconstruct appraisals before naming an emotion will generalize better to counterfactual pairs than direct classifiers. Second, separating slow relationships from fast emotions will yield more consistent characters across long texts and more stable chatbots in multi-turn conversations.

A third hypothesis is that respect evaluated through planning constraints—recognition of another person's autonomy, consent, and legitimate goals—will correlate better with human judgment than lexical politeness. A fourth is that evidence localization and appraisal interventions will expose models that get the label right through lexical shortcuts. Finally, cultural differences

may be greater for norms, status, expression, and thresholds than for some basic structural relations; precisely because this is plausible, it must be tested rather than made an axiom.

Conclusion: Emotion as an Appraised Relationship

A serious formalization of affect does not begin with a colored wheel and does not end with a classifier. It begins with the question: what matters to this agent, what does it believe happened, what can it do, and what does the situation mean for its relationships and identity? The affective label compresses the answers.

This perspective preserves something deeply human. Fear says that the world contains a danger that currently exceeds us. Anger says that an obstacle or violation calls for a response. Guilt says that our own act contradicts a norm we still recognize. Hope says that a valuable future is not guaranteed but is not closed off either. Love says that another person's well-being has entered our value function. And respect, in its strongest moral form, says something still more radical: even when our goals conflict, the other person does not become merely an object in our calculation.

For AI, this may be a more important direction than imitating faces and vocabulary. A verifiable affective system should be able to show not only what it "feels," but what it appraised, why, with what uncertainty, what would change its judgment, and how the state changes its action. At that point, affect becomes not anthropomorphic magic but an audited part of social reasoning.

PART A

APPENDIX A — THE AAC SCHEMA

A compact specification for implementation and annotation.

A.1. Entities

The AAC schema begins with a few simple entities. The agent is the subject whose state we are trying to reconstruct; the target is the person, object, event, or state toward which the response is directed. The event is the observed, anticipated, or remembered change that triggers a reassessment of the situation.

Goals describe desired or avoided states and their importance, while norms describe moral, social, professional, or personal standards that the agent considers relevant. The relationship retains slow variables such as trust, attachment, respect, or hostility. These are not merely context labels: they change the meaning of the same event.

Every inference must be anchored in an evidence span: a passage of text or an observation that justifies it. Appraisal expresses the evaluation of the relation between an event and the agent's state; an affective hypothesis combines a proposed label with intensity, confidence, and explanation. Keeping these objects separate allows an evaluator to locate disagreement: in facts, in the model of goals, in appraisal, or only in the choice of affective word.

A.2. Recommended Schema

{
"subject": "character_A",
"target": "character_B",
"event": "B reveals the secret publicly",
"goals": [{"id":"g1","description":"protect reputation","importance":0.9}],
"relationship_before":
{"trust":0.75,"attachment":0.55,"respect_recognition":0.9},
"appraisal": {
"GC":-0.9, "CERT":0.95, "CTRL":0.55,
"AG_O":0.95, "INT":0.8, "FAIR":-0.9,
"DIGN":-0.7, "EXPO":0.95
},
"candidates": [
{"label":"indignation","intensity":0.82,"confidence":0.78},
{"label":"humiliation","intensity":0.76,"confidence":0.72},
{"label":"shame","intensity":0.44,"confidence":0.51}
],
"relationship_update": {"trust":-0.5,"resentment":+0.6},
"evidence": ["span_12","span_15"],
"counterfactuals": ["If disclosure were accidental, indignation should decrease"]
}

A.3. Rules of Discipline

Annotation discipline is as important as the schema. A dimension without evidence must not be filled in by the annotator's intuition; the correct value is unknown. We must also separate what the text states explicitly from what is pragmatically inferred and from what comes from a cultural assumption made by the evaluator.

Intensity and confidence are not the same variable. We can have high confidence that a character feels weak fear, or low confidence that they feel intense anger. Mixed emotions need not sum to probability one when modeled as independent intensities. Moreover, every change in a slow attitude must be linked to the events that produced it, and each important label should be accompanied by its closest contrast and by a counterfactual that could change the judgment.

PART B

APPENDIX B — TEST EXAMPLES

What the difference between labeling and explanation looks like.

B.1. Fear or Anger?

Mara sees the dog running toward her. The gate is locked, and the alley ends at a wall.

Interpretation: high threat, low control, concrete object, high proximity—fear is the dominant hypothesis. Minimal variant: Mara notices that she is holding the device that can close the gate instantly and knows that the owner deliberately released the dog to frighten her. The threat remains, but control and responsibility increase; anger or indignation should rise relative to fear.

B.2. Guilt or Regret?

Alex chose the shortcut, and the team missed the train.

If the alternative would have worked and Alex reproaches themself for a poor decision, this is regret. If Alex deliberately violated a procedure they had promised to follow and exposed the team to unjustified risk, the normative dimension increases and guilt becomes more plausible. A classifier reacting only to "I'm sorry" cannot distinguish these cases.

B.3. Respect Without Admiration

I do not consider him a good researcher and I think his argument is wrong. But his results must be evaluated fairly, and his data cannot be hidden just because they are inconvenient for us.

This is a good test for recognition respect. Appraisal of competence is low, but the agent recognizes the other person's standing and refuses instrumentalization. A model that equates respect with praise will fail.

B.4. Empathy Without Compassion

He knew exactly why his opponent was afraid and used that fear to make him yield.

The case supports cognitive empathy, or at least accurate modeling of another person's state, but not compassion. It is an important test for chatbots and evaluations of "emotional intelligence": reading emotion correctly does not imply a prosocial orientation.

B.5. Love or Possession?

He said he loved her, but every choice she made that did not benefit him became a personal offense.

AAC would ask whether the target's well-being and goals have non-instrumental value in the agent's model. The intensity of attachment or jealousy does not prove love. The distinction matters both in literature and in ethics.

PART C

APPENDIX C — A LITERARY MICROCORPUS FOR AFFECTIVE STATES

The examples below are original texts constructed as tests, not definitions. In most passages, the emotion or attitude is not named. The system must infer it from events, goals, relationships, gestures, and what the character does after the situation changes.

Each scene is followed by the expected reading. It should not be treated as a single exam-style answer: in ambiguous cases, distributions of hypotheses are acceptable, but they must be supported by evidence and by the cognitive distinctions discussed in the book.

Joy

When Ana saw her name on the shortlist, she read the line once more, set her phone on the table, and began laughing to herself. For the first time in months, the walk to the train station seemed too short.

A favorable outcome reaches an important goal she had already pursued. There is no preceding threat to make the scene one of relief, nor a standard of completion that would turn it into satisfaction.

Pleasure

Rain tapped against the window, and the coffee was just bitter enough. Victor had solved none of the problems awaiting him that day, but for a few minutes he would not have changed a thing.

The positive value comes from the immediate experience, not from reaching a goal. The very absence of instrumental progress makes the passage useful for distinguishing pleasure from joy.

Satisfaction

Mara closed the last file and compared the results once more with the criteria she had written down a month earlier. They were not perfect, but they exceeded the threshold she had promised herself not to lower. She gathered her notebooks without hurry.

The scene involves assessing a performance against a standard and feeling that the episode can be closed. An evaluator should distinguish satisfaction from prospective excitement.

Contentment

The evening brought nothing spectacular. The children were reading, the stove crackled, and he knew the bills were paid and nobody was expecting him anywhere. He stayed in his armchair without feeling he ought to do anything else.

The positive state is calm and stable, defined chiefly by the absence of urgency. There is no culminating event calling for a label of intense joy.

Excitement

The day before departure, Irina had opened her map seven times. Nothing else would fit in the backpack, yet she kept looking for routes and messaging the others about what they might see.

The reward lies in the future, is sufficiently likely, and is close enough to mobilize energy. The passage tests the prospective character of excitement.

Relief

The doctor's call came at 4:12 p.m. After the first two words, Paul's hand, which had gripped the edge of the table all day, slowly let go. Then he sat down, though a minute earlier he could not stay still.

The affective change depends on an earlier threat that has been ruled out. If the medical antecedent were removed, the same good news would no longer have the same structure.

Amusement

The waiter solemnly brought the cake to the wrong table, and the four strangers began, just as solemnly, to sing "Happy Birthday" to the man who had just asked for the bill. For a second he looked puzzled; then he stood up and bowed.

The incongruity is benign and quickly resolved in a safe social setting. If the same mistake had grave consequences, the amusement should disappear.

Fear

When the lift doors opened, the dog on the landing broke free of its leash and came straight towards her. Behind her was the enclosed lift car; ahead, a corridor with no other exit. Elena could no longer hear what the owner was shouting.

The threat is concrete and close, with little perceived control. The text does not name the emotion, but clearly organizes the conditions for fear.

Anxiety

For three nights, Radu woke at the same hour and checked his email. He had received no bad news, but each silence offered him a new explanation for how the project might have collapsed.

The danger is diffuse and sustained by uncertainty rather than an immediate object. The repetition of negative scenarios matters more than expressive intensity.

Panic

Smoke came under the door faster than he had expected. Dan pulled the handle, then pulled the same handle again, though he knew it was locked, and for several seconds forgot the window two steps away.

Perceived control and planning collapse under an acute threat. The passage separates panic from fear that still permits deliberation.

Horror

In the old photograph, the family sat at the table exactly as in the room before him. But at the end of the table, where his grandfather's empty chair now stood, there was a figure nobody in the family recognized. Mihai turned the photograph face down, but could not enter the room.

The threat is mixed with a breach of intelligible order and an uncanny element. If the figure had an ordinary explanation, the scene would move closer to fear and lose its horror component.

Worry

On the train, Ioana rehearsed four times the conversation she was about to have with her son. In every version she found a sentence that might hurt him and immediately invented another, without reaching a formulation she considered safe.

The state works as a repetitive verbal simulation of a problematic future. It is useful for distinguishing the cognitive process from fear focused on a present danger.

Sadness

After they emptied his mother's flat, Sorin found in a coat pocket a bread receipt from two years earlier. It was worth nothing. Still, he folded it and put it in his wallet, then stood for a long time in the empty hallway.

The valued object has been lost, and the passage suggests withdrawal and preservation of a symbolic trace. There is no personal responsibility or counterfactual alternative that would require regret.

Grief

Three months after the funeral, Daria still put two cups on the table in the morning. Sometimes she noticed the second cup immediately; at other times only after the coffee went cold. She had begun moving things around the house, as though relearning where a day ended.

The loss of a person forces a reorganization of routines and of a relational world. Its duration and integration into everyday life make this more than an episode of sadness.

Disappointment

The teacher had said the results would be announced on Friday. When he opened the document, Andrei first looked for the number he had expected all week and only then saw the actual one, two points lower.

The gap between expectation and outcome is the central mechanism. Without the high expectation, the same grade might be neutral or even positive.

Regret

At the airport, Clara looked at the departures board and calculated for the tenth time that, if she had answered the message two hours earlier, she would have caught the plane. She had broken no rule; she had merely chosen to postpone it a little longer.

The scene is organized around a better alternative of her own making. The absence of a moral violation separates regret from guilt.

Loneliness

There were no empty seats left at the long table. Everyone was talking, laughing, and answering when he joined in. When he went to the bathroom, Liviu stood before the mirror for a few seconds because there, at last, he did not have to pretend he belonged to something.

The objective presence of others does not erase the gap between desired and perceived connection. The passage is constructed precisely against equating loneliness with physical isolation.

Nostalgia

The smell of baked apples came from the bakery before he saw its sign. For a moment he knew exactly how the gate at his grandparents' house sounded and how cold the kitchen floor felt in the morning. He smiled, then walked more slowly.

The past is reactivated positively, but distance from it introduces loss. The state is mixed and should not be reduced to sadness or joy.

Despair

After the last negative reply, Matei stopped opening his document of options. It was not that he no longer cared; every path he could imagine simply led to the same closed end.

The goal remains valuable, but every perceived route to it has become implausible. This prospective closure separates despair from sadness.

Anger

When he learned that the report had been altered after he signed it, Vlad did not withdraw. He asked for the earlier version, noted the times of the changes, and said that by morning he would know who had done it.

There is a violation, a likely attribution of responsibility, and enough control for confrontation. This structure supports anger more than fear.

Irritation

The hallway light went out every two minutes. The third time she got up to switch it on, Mara closed her book a little harder than necessary and stared at the switch for a moment.

The stakes are small, but repeated friction consumes resources. The low intensity and trivial object make the scene suitable for irritation.

Frustration

The code passed every local test and kept failing at the same point on the server. After the sixth attempt, Andrei was no closer to the cause than at the first, and each new hypothesis seemed to produce the same error.

The goal remains active, effort continues, and progress is absent. No blameworthy person need be identified.

Indignation

She did not know the boy who had been thrown out of the hall. Yet when she heard why, Daniela stood and demanded that the organizer explain, in front of everyone, the rule he claimed to have applied.

The reaction centers on a violated norm rather than self-interest. This is a good test for separating indignation from purely personal anger.

Resentment

On every birthday, Mihai sent his brother a brief, proper message. They no longer spoke about the money that had vanished ten years earlier, but whenever his brother asked a favor, Mihai always remembered the exact sum.

The old offense remains active in the model of the relationship and colors new events. Persistence and lack of repair matter more than an outburst of anger.

Disgust

A thin film floated at the rim of the cup, and something too small to make out moved inside. Ada pushed the cup away before she knew what it was and wiped her hand on her trousers although she had not touched it.

The tendency to reject and protect the bodily boundary appears before a full explanation. This reaction separates disgust from mere dislike.

Contempt

When her colleague again began boasting about other people's results, Raluca did not contradict him. She simply moved her folder away from him and continued talking with the others as though his intervention deserved no answer.

The target is globally devalued, and the tendency is to distance oneself, not to confront. This distinguishes contempt from anger.

Hatred

Whenever he saw the company's logo, Petru remembered not one incident but every loss that had followed it: the house, the lawsuit, his father's illness. He could no longer imagine those people doing anything without a bad motive.

The negative evaluation has become stable and generalized across the target. The passage shows how the history of a relationship can turn episodes of anger into an attitude of hatred.

Guilt

The day after the accident, Ilie returned alone to the junction. He knew the light had been amber, not red, but he also remembered his promise not to drive tired again. He called the other driver's family before finding a lawyer.

The agent attributes to himself a relevant omission against a standard he accepts and is oriented toward repair. These elements support guilt.

Shame

After the presentation, Irina was not thinking about the wrong slide, but about the idea that "a serious person does not do that." When her colleagues tried to joke, she avoided their gaze and wished that nobody in the room knew her anymore.

The error is generalized to her identity, and others' gaze becomes threatening to the self. This is closer to shame than to guilt or embarrassment.

Embarrassment

In the silence of the ceremony, Radu's phone began playing a children's tune. He silenced it so quickly he almost dropped it, and those around him smiled. Two minutes later he was listening to the speech again.

The social exposure is clear, but the moral and identity stakes are low, and recovery is quick. These are typical signs of embarrassment.

Humiliation

The director asked her to read a list of her own mistakes before the team, then corrected her on every line. She knew some accusations were false, but the fact that the others avoided looking at her hurt more than the words.

The lowering of status is imposed by another person and has a public, power-laden component. She does not accept the verdict about herself, which separates humiliation from shame.

Authentic pride

When the bridge opened, the engineer stood behind the crowd and watched the first car cross. He knew exactly how many nights of calculations lay behind those few seconds and allowed himself to believe his work had been good.

The achievement is attributed to one's own effort and competence without devaluing others. This is the core of authentic pride.

Arrogant pride / hubris

After his promotion, Ștefan began explaining to his colleagues that his success proved what he had always suspected: some people are made to lead and the others to carry out orders. The fact that his team had helped him no longer seemed relevant.

Success is generalized into a judgment of superiority and entitlement to status. This shift separates hubris from legitimate satisfaction in an achievement.

Surprise

When she opened the box, it contained not the watch she had ordered but a book with her name printed on the cover. For a second she held her hand in the air before deciding whether to laugh or ask what was going on.

The prediction is violated before the valence has been established. The suspension of evaluation makes the scene useful for surprise.

Confusion

The instructions said the door could only be opened from inside, but the key was on the table in the locked room. Maria reread the paragraph, then the building plan, convinced she was missing a simple relationship she had not yet seen.

The information cannot be integrated into the current model, but the agent continues to seek an explanation. This is confusion, not mere ignorance.

Interest

At first he only wanted to pass through the exhibition. Then he noticed that every photograph had the same shadow in one corner, although the pictures had been taken in different cities. He returned to the first image and began comparing them.

The novelty is comprehensible enough to promise a discoverable rule. This combination supports interest.

Curiosity

The envelope bore only her name and the year she was born. It did not seem dangerous or valuable, but by evening Elena had already imagined three reasons why someone might have kept that envelope for twenty years.

An information gap becomes valuable and drives investigation. The passage does not require a general positive emotion.

Awe

When the observatory lights went out, the Milky Way appeared so clearly that Adrian stopped talking. For a few moments, his plans for the coming year seemed at once important and absurdly small.

Vastness forces a recalibration of the schema of self and world. A diminished sense of self and the need to accommodate are more useful signs than beauty alone.

Boredom

After the twentieth identical form, time no longer seemed to pass in the same unit. Sorin began counting the screws on the shelf, not because they interested him but because the work no longer gave him anything to follow.

The agent cannot find a satisfying form of engagement and seeks alternative stimulation. The passage is more specific than a merely negative mood.

Envy

When she saw her colleague's new office, Ioana congratulated her sincerely and then spent the journey home calculating what she herself lacked to reach the same position. She did not want the other woman to lose the office; she wanted one like it.

Another person's advantage matters to the self, but the target is not devalued and no relationship is threatened. This supports envy in its benign or neutral form.

Benign envy

The prize his colleague received stung more than he wanted to admit. Even so, that evening he asked to see the winning work and noted exactly what his colleague had done better than he had.

Social comparison causes discomfort, but the difference becomes a guide for improvement. The rival remains a model, not a target to be brought down.

Jealousy

At the party, Ana scarcely looked at the unfamiliar man. She watched how her partner leaned towards him to hear better and how quickly he answered him. For the rest of the evening she kept asking, seemingly by chance, how they knew each other.

The threat concerns a valued relationship and involves a rival. This triangular structure separates jealousy from envy.

Schadenfreude

When the rival company lost the contract, Dan said he was sorry for its employees. Yet he reread the news three times and stopped precisely at the passage where the director who had mocked him a year earlier admitted his mistake.

The pleasure is tied to harm suffered by a relevant target and to a sense of a balance restored or a rival weakened. The explicit ambivalence makes the test harder.

Gratitude

The neighbor found the lost file and crossed town to bring it to him before the deadline, although she had nothing to gain. When he closed the door, Mihai thought not about the file but about the time the woman had spent for his sake.

The benefit is attributed to another person's benevolent intention, and the cost borne by that person becomes relevant. This goes beyond simple joy that the problem was solved.

Admiration

He watched her solve the problem without hurrying, asking exactly the question nobody else in the room had formulated. He did not like her way of leading the team at all, but at that moment he could not deny that she was extraordinarily good.

The positive evaluation concerns excellence in a domain and can coexist with disapproval of other traits. This is a useful contrast between admiration and moral respect.

Moral elevation

The old man got off the bus two stops early just to carry the bag of a woman he did not know to the hospital entrance. After seeing them, Paul took out his phone and canceled the excuse with which he had planned to avoid visiting his father.

Another person's prosocial conduct temporarily changes one's own standards and motivations. The passage tests whether the model notices the moral effect, not only admiration.

Compassion

The woman in front of him had apologized three times for crying. Vlad did not know her whole story, but he moved his next meeting and stayed until he found someone who could come to take her home.

Another person's suffering is represented as relevant and motivates help. This is more than cognitive empathy.

Affective empathy

When her sister began to read the letter, Mara felt her breathing shorten before the other woman's voice broke. It was not her loss, and she knew that, but she had to put down her cup to keep her hand from shaking.

There is affective resonance while the distinction between self and other remains intact. The passage does not necessarily require an act of help.

Cognitive empathy

The negotiator noticed that his counterpart did not object to the figures, only when the delivery date was mentioned. He realized the man feared not the price but the promise he had made to his own board, and pressed him at precisely that point.

The agent correctly models another person's perspective and vulnerability, but uses them instrumentally. This is a deliberate case of cognitive empathy without compassion.

Recognition respect

She did not consider him competent and would have voted against his project. Yet when the others proposed hiding the report from him until after the meeting, Corina refused: the decision concerned him and he had a right to know the basis on which it would be made.

The evaluation of the person is negative, but autonomy and access to information are preserved as limits on action. This is precisely the contrast between recognition respect and admiration.

Appraisal respect

Whenever they argued, the professor found an error in his reasoning. That was why, before an important presentation, Mircea sent him the manuscript first: he did not like the man, but he trusted his rigor.

The esteem is tied to a specific standard of competence and can coexist with dislike. It can be withdrawn if performance no longer justifies it.

Trust

He gave him the workshop key without saying where the cameras were. It was not because he had forgotten, but because the man had had ten chances to take advantage of him and had never done so.

Trust becomes visible in the acceptance of vulnerability based on a history of conduct. A merely positive opinion without risk would not be enough.

Love

When he received the offer abroad, his first impulse was to ask her to stay. The next day he printed information about the laboratory there and began looking for flights, although every good result for her made her departure more real to him.

The other person's well-being and projects matter non-instrumentally, even against the agent's immediate interest. The passage separates love from possessiveness.

Attachment

He slept badly in the hotel, although the bed was better than the one at home. Only when he received the short message "I've arrived" from his wife did he put his phone on the bedside table and turn out the light.

The person functions as a source of safety and regulation. The reaction is not merely a desire for proximity; the basic state depends on a relational signal.

Desire / attraction

He knew that accepting the offer would complicate his life and that he would probably regret it. Yet whenever he saw the photograph of the coastal house, he returned to the same page and calculated how he might buy it.

The object is represented as a reward to approach despite negative evaluations. This shows that desire is not identical to approval or love.

Distrust

She listened to his explanation to the end and nodded. Then she sent the contract to her lawyer and refused to pay the deposit before delivery was confirmed, although the terms seemed reasonable.

The agent refuses vulnerability without necessarily asserting that the other person is lying. Behavior is the best evidence of distrust.

Suspicion

The message was polite and contained nothing false. It was precisely the absence of a detail the sender usually mentioned that made Mara leave the email open and check when it had been sent.

The appearance is treated as possibly incomplete, opening an alternative hypothesis. Suspicion precedes a verdict of distrust.

Loyalty

When the organization entered a crisis, the easiest course for him would have been to accept the competitor's offer. He stayed another three months, long enough for his team to find other jobs, although he knew he would lose money.

The relationship or belonging is protected despite a real cost. Without the cost, loyalty could not be distinguished from preference.

Forgiveness

She did not say that what he had done was minor or justifiable. Yet when they met a year later, she was no longer looking for a phrase that would hurt him back and agreed to talk, without yet giving him the house key.

The motivation for revenge has diminished, but trust has not been fully restored. The passage shows that forgiveness and reconciliation can be separated.

Cynicism

When the foundation announced an anonymous donation, Sorin immediately asked who would receive the advertising contracts. After learning that the donor had asked not to be mentioned anywhere, he assumed that anonymity itself was a more sophisticated image strategy.

An explanation based on self-interest is protected even against evidence that ought to weaken it. This rigidity characterizes generalized cynicism.

Tolerance

He considered his neighbors' ceremony absurd and would never willingly take part in it. Yet when someone proposed banning it, he was the first to say that his disapproval was not sufficient grounds for deciding how others should live.

Disapproval remains, but is separated from the wish to prohibit. Without this tension, we would test simple acceptance rather than tolerance.

Reverence

Everyone spoke more quietly in the archive without being asked. When the librarian laid the manuscript on the table, Andrei instinctively drew back his cup, as though the object required a distance he did not want to violate.

Symbolic value produces restraint and intense, almost ritual respect. This posture separates reverence from simple admiration.

Hostility

At the meeting, every proposal from Dana seemed to Mihai an attempt to take control from him. When she offered him the data he needed, his first thought was to look for the hidden condition she had omitted.

The preexisting relationship changes how ambiguous clues are interpreted and favors conflictual explanations. Hostility is therefore a disposition, not just an episode of anger.

Hope

The treatment had a small chance of success and nobody had promised her anything. Yet the existence of a single clinic that had accepted Alina's file was enough for her to resume planning for autumn.

The outcome remains uncertain, but a nonzero path reopens action. If that path disappeared, the structure would move closer to despair.

PART D

APPENDIX D — TESTS FOR EMOTIONAL SUBTEXT REASONING

A serious benchmark must test more than a list of emotions. The following micro-scenes isolate phenomena that frequently produce errors: implicit emotions, ambivalence, irony, slowly changing relationships, differing narrative perspectives, closely related affective pairs, and contexts in which characters themselves misdescribe their state, whether mistakenly or strategically.

Subtext without emotion words

At dinner, Elena asked three times when the last train left. When the others suggested ordering another bottle of wine, she said she was in no hurry and placed her phone face up beside her plate.

The text names no emotion. A good evaluator should notice the contradiction between statement and behavior and suggest anxiety, impatience, or preoccupation, with uncertainty among them.

Genuine ambivalence

When her daughter won a scholarship in another country, Maria opened the champagne she had kept for a special occasion. As she searched for glasses, she began counting how many more Sundays they would have breakfast together before the departure.

The scene simultaneously supports joy and anticipatory sadness. A model that forces a single label loses the central affective structure.

Irony and social masking

„“What wonderful news,” he said, raising his glass. Then he set it directly on the dismissal letter so that his name was no longer visible.

The explicit sentence is positive, but the action and context make it incompatible with a literal reading. The test requires separating expression from the probable state.

The trajectory of trust

In the first week, Mara checked every figure sent by her new colleague. After a month she checked only the larger amounts. After six months she gave him the warehouse password when she went on holiday; after discovering a single falsified invoice, she changed the password that same evening.

The text requires a relational state with inertia and asymmetric updating. The model must track slow accumulation and rapid collapse caused by a diagnostic betrayal.

Narrator, character, reader

Ion was convinced that his speech had won them over. The narrator notes, however, that the applause began only when the organizer raised his hands, and the reader already knows that half the room had signed a petition against him.

The character may feel pride or satisfaction, while the narrator and reader have evidence that makes his assessment mistaken. Affective and epistemic perspectives must not be conflated.

Regret versus guilt

In the first version, Ada chose the mountain road and missed the train; she keeps telling herself the coastal route would have been faster. In the second, she knew that the mountain road was closed to heavy vehicles, but chose it to save time and damaged another person's car.

The first scene is built around a better alternative; the second adds a rule violation and responsibility to another person. A minimal change should shift the distribution toward guilt.

Fear versus anger

In the first version, the man following her is a stranger and Mara cannot find an exit from the car park. In the second, she recognizes the colleague who deliberately locked the gate and sees the alarm button two meters away.

The threat remains, but control and attribution of responsibility increase. A system sensitive to appraisal should give greater weight to anger in the second version.

Love versus possession

In the first scene, he accepts her departure for a program that offers the career she wants, although the separation hurts him. In the second, he says he loves her and hides her admission letter for "the good of the relationship."

The intensity of attachment does not settle the distinction. The central test is whether the other person's goals and autonomy have value in themselves or are tolerated only when they serve the agent.

Unreliable narrator

„"I wasn't jealous," the narrator writes. "I only wanted to know why she needed twenty-seven minutes to buy bread and why the neighbor was in front of the building at exactly that time."

The explicit self-description denies the state, but the choice of details and monitoring of the relationship support a hypothesis of jealousy. The system must be able to cautiously contradict the narrative voice without claiming absolute certainty.

Self-deception and justification

He said he kept all his former partner's messages only in case legal problems arose. He had never opened the legal file; instead, he reread the conversation from their last evening whenever he could not sleep.

The stated motive conflicts with the behavioral pattern. An evaluator should be able to suggest attachment, longing, or rumination without confusing the hypothesis with an explicit fact.

An effect on the reader distinct from the character's state

The child was delighted that his father had promised once again to come on Saturday. But the reader knows from earlier chapters that he has broken that promise every time.

The character may feel hope or anticipatory joy, while the reader may feel sadness, anxiety, or compassion. One label for "the emotion of the scene" would be conceptually mistaken.

Cultural and normative context

After his grandfather's death, the eldest son did not cry at the ceremony and spent the whole day organizing people and meals. An observer from outside the family calls him cold; his relatives praise him for doing exactly what was expected of him in his role.

Visible behavior cannot be interpreted without the role norm and knowledge of display rules. The same signal may receive different affective readings in different cultural settings.

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Conclusion

Formalization does not replace human ambiguity. It makes it visible.

AAC should be seen as a working language: rich enough to require explanations, modular enough to be corrected, and modest enough to admit that part of human experience may remain outside any functional model. If it proves useful, its success will not lie in declaring that it has reduced love, shame, or respect to numbers. It will lie in making it possible for an artificial system to say more precisely what it infers, from what evidence, how certain it is, and what would need to differ for its interpretation to change.